TechGap: An Intelligent Decision Support System for Syllabus-Industry and Career Roadmapping for CICS Students of De La Salle University – Dasmariñas Using Job Market Analytics

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Chapter I

INTRODUCTION

1.1 Project Context

The rapid development of technology has continuously changed the skills and competencies required across industries, creating a need for higher education institutions to keep their academic programs responsive to evolving professional requirements. As technologies develop and industries adopt new tools, systems, and practices, the competencies expected from graduates also continue to change. This creates an ongoing challenge for universities in maintaining alignment between the knowledge and skills developed through their academic programs and the requirements of the labor market. Academic programs may face difficulties in keeping pace with current technological advancements and job-market demands, which can result in gaps between what graduates know and what employers require (Karakolis et al., 2022). In computing-related programs such as Computer Science and Information Technology, where technological developments directly influence professional practices, maintaining this alignment requires continuous attention to changes in the field.

In the Philippines, the development of computing programs in higher education is guided by national policies, standards, and guidelines established by the Commission on Higher Education (CHED). CHED Memorandum Order No. 25, Series of 2015 provides the Revised Policies, Standards, and Guidelines for Bachelor of Science in

Computer Science (BSCS), Bachelor of Science in Information Systems (BSIS), and Bachelor of Science in Information Technology (BSIT) programs. The memorandum adopts an outcomes-based approach and establishes minimum standards through prescribed program and graduate outcomes. At the same time, it recognizes the rapidly changing nature of computing and allows higher education institutions to develop and improve their curricula according to their institutional context and academic objectives while maintaining the required learning outcomes (Commission on Higher Education [CHED], 2015). The framework therefore provides a national basis for computing curriculum development while allowing institutions to determine how their programs can be structured and maintained within their respective contexts. As these programs are expected to remain responsive to developments in computing, their relevance is also shaped by the changing conditions of the technology sector and the skills required within it.

Recent Philippine labor statistics provide a view of the employment environment in which these computing programs operate. In 2025, the Philippine digital economy generated approximately ₱2.74 trillion in gross value added, equivalent to 9.8 percent of the country's gross domestic product, and employed approximately 10.39 million persons, representing 21.2 percent of total Philippine employment (Philippine Statistics Authority [PSA], 2026b). The digital economy covers a broad range of activities beyond computing professions, including digital-enabling infrastructure, digital content and media, e-commerce, and government digital services (PSA, 2026b). A narrower view of technology-related employment is provided by the Information and Communication sector. In 2024, the sector recorded 2,454 formal establishments and employed 172,389

workers. Computer programming, consultancy, and related activities accounted for 90,573 workers, or 52.5 percent of employment within the sector, making it the industry group with the largest number of workers (PSA, 2026a). These statistics describe both the broader scale of digital economic activity and the more specific employment setting of programming and related computing activities in the Philippines.

Alongside this employment context, current Philippine technology job-market data provide information on the occupations and technical requirements appearing in advertised positions. Get U Hired's 2026 Philippine technology labor-market analysis currently covers 21,789 unique public job postings collected from June 30 to September 28, 2026, with an average of 1,676 new listings per week (Get U Hired, 2026). Within the analyzed postings, Backend Engineer was among the most frequently advertised technology roles, while SQL, Python, and Microsoft Azure were identified as the leading requested technical skills and technologies (Get U Hired, 2026). These advertised requirements cover areas such as programming, databases, and cloud technologies. Beyond individual job advertisements, the Philippine Skills Framework for Software Development and Security identifies employment opportunities and competency areas across operations and support, product development, cybersecurity, and governance, risk, and compliance. Developed through consultation and validation involving Philippine employers, education and training providers, practitioners, government, and academe, the framework describes technical competencies associated with software development and security within the Philippine setting (Department of Information and Communications Technology [DICT] & Philippine Software Industry Association [PSIA], 2024).

Within this national employment and industry context, De La Salle University–Dasmariñas (DLSU-D), particularly through the College of Information and Computer Studies (CICS), develops its undergraduate computing programs within the standards established by CHED. The BSCS and BSIT programs follow the national framework provided by CHED Memorandum Order No. 25, Series of 2015 while being structured according to the institution's academic context and objectives. These programs provide students with the academic foundation necessary for developing computing competencies, while their continued relevance depends on the ability to evaluate whether the competencies represented in the curriculum remain responsive to developments in the technology sector.

However, continuously examining curriculum content against changing industry requirements can be challenging when curriculum information and large volumes of labor-market data must be reviewed systematically. Traditional curriculum evaluation remains dependent on the knowledge and judgment of faculty members, program administrators, industry representatives, alumni, and other academic stakeholders. While this human judgment remains essential, manually examining course content alongside a continuously changing body of industry and job-market information can require considerable time and effort. Curriculum mapping research has identified curriculum analysis as potentially costly and time-consuming, while studies of online job advertisements have demonstrated the need for systematic methods to extract and organize skills from large volumes of job-market text (Agahi & Salwén, 2025; Khaouja et al., 2021).

This creates a need for a more systematic approach that can organize curriculum information and industry-derived skill data in a form that supports curriculum evaluation. In response to this need, the study proposes TechGap, an intelligent decision support system designed to extract technical skills from course syllabi and compare them with technical skills derived from job-market data. Through this comparison, the system is intended to identify areas of alignment, potential skill gaps, and possible mismatches between competencies represented in the curriculum and those reflected in current industry requirements.

TechGap is intended to serve as a decision-support tool rather than a replacement for the judgment of curriculum developers, faculty members, and program administrators. By organizing curriculum and industry skill information into structured evidence, the system can support academic stakeholders in examining areas that may require further evaluation or strengthening. In this way, the project provides a systematic means of examining curriculum-industry alignment while maintaining the institutional and national standards that guide computing education in the Philippines.

1.2 Purpose and Description

The purpose of this study is to develop the Curriculum Gap Analyzer, a platform designed to help De La Salle University-Dasmariñas (DLSU-D) assess the academic programs and transform the implemented curricula through data-driven, strategic alignment with evolving industry standards. The main goal of this study is to identify possible gaps between the skills taught in university courses and the skills required by the industries to ensure that College of Information and Computer Studies (CICS) graduates are better prepared for employment, as well as establishing the system as an Intelligent Decision Support System (IDSS) that leverages predicting modeling and prescriptive analytics to optimize curriculum design and implementation.

The system functions as a proactive tool for curriculum optimization and institutional decision-making. It will work by continuously integrating course curricula and real-time job market data to analyze them, enabling the academic stakeholders to anticipate emerging skill demands and evaluate strategic curriculum adjustments prior to formal adoption. Through this function, an embedded simulation is implemented to allow the administrators to model and project the potential outcomes of proposed curriculum modifications, supporting data-and evidence-based planning by providing an iterative and predictive process of curriculum evaluation and refinement.

Moreover, the innovative aspect of the system lies in its deterministic recommendation engine, which generates structured, explainable outputs grounded in

defined rules and academic constraints. Whereas all recommendations are checked against national standards, such as minimum unit requirements and preservation of mandated core courses, to ensure maintenance with institutional compliance. Therefore, the system suggests curriculum revisions without violating any academic policies. By using data analysis and structured validation, the system provides recommendations that are relevant, practical, and suitable for implementation.

Furthermore, the platform provides dual-purpose outputs designed to address both administrative and student needs. For academic administrators and stakeholders, the system generates prescriptive revision guides that outline prioritized curriculum enhancements, course-level modifications, and projected improvements in alignment. And for students, it provides skill roadmaps that link to learning resources based on the technological career they want to enter after graduation, thereby supporting the students in independently bridging the gaps between the current curriculum and industry-required skills. These skill roadmaps guide students in identifying competencies that may not be fully covered in their courses and encourage self-directed learning through recommended resources. This will help students be more prepared, competitive, and aligned with industry expectations even before graduation.

By using this tool, the university can continuously evaluate and refine its curriculum to remain aligned with current and emerging industry demands. This enables timely, data-driven updates that reduce the gap between academic preparation and

workforce requirements. As a result, graduates are better equipped with relevant skills, making them more prepared and competitive in both local and global job markets.

1.3 Statement of the Problem

The curriculum currently utilized by the College of Information and Computer Studies (CICS) is based on 2019 standards. In the technology sector, a five-year-old framework may become less responsive to current industry requirements as tools, roles, and technical competencies continue to evolve. Although curriculum reviews are conducted periodically, manual and incremental updates may not fully capture real-time labor market demands.

This creates a need for a data-driven system that can assess curriculum–industry alignment and support academic stakeholders in identifying competency gaps, low-demand topics, and possible curriculum improvements. At the same time, students also need a way to understand how their completed or mastered courses align with target career roles, what skills they already possess, and what competencies they still need to develop.

Specifically, this study seeks to answer the following:

1. How can TechGap extract and process technical skills from CICS curricula and Philippine job market data to support curriculum–industry alignment analysis?
2. How can TechGap identify competency gaps, logically covered skills, and low-demand curriculum topics through semantic matching and inference-based analysis?
3. How can TechGap generate curriculum recommendations and summary reports that support academic administrators and faculty in curriculum review?

4. How can TechGap help students assess their personal alignment between completed or mastered courses and target career roles, including their skill gaps and learning roadmap?
5. How acceptable is the system in terms of functionality, usability, accuracy, efficiency, and reliability?

1.4 Hypothesis

In this study, the independent variable is the use of the TechGap system in curriculum evaluation and career preparation by College of Information and Computer Studies (CICS) students and academic stakeholders. The dependent variables include both system performance metrics, namely functional accuracy, reliability, and usability, and outcome-based measures such as perceived usefulness, decision-making support, career alignment clarity, and curriculum alignment score. These variables are examined to assess not only the technical performance of the system but also its effectiveness in supporting users in understanding curriculum–industry alignment and enhancing their readiness for academic and professional development. The null hypothesis states that the use of the TechGap system has no significant effect on both system performance metrics and user-centered outcomes, while the alternative hypothesis states that the use of the TechGap system has a significant level of effectiveness across these measures as evaluated by CICS students and academic stakeholders.

1.5 Research Objectives

1.5.1 General Objective

To design and develop TechGap, an intelligent web-based Decision Support System that utilizes natural language processing and semantic similarity modeling to identify skill gaps between CICS academic curricula and Philippine job market requirements, while also helping students assess their personal curriculum-to-career alignment.

1.5.2 Specific Objectives

1. To identify in-demand skills from job postings by developing a data acquisition and skill extraction pipeline that processes Philippine job market data.

2. To analyze the alignment between curriculum competencies and industry requirements by implementing semantic matching, skill ontology-based inference, and adaptive ranking mechanisms. s.

3. To detect skill gaps in the current curriculum by designing both an admin/faculty dashboard for curriculum recommendations and a student-facing module for personal career roadmapping.

4. To evaluate the system using ISO 25010 standards to establish a structured baseline for software quality assessment.

5. To evaluate the acceptability of TechGap in terms of functionality, usability, accuracy, efficiency, and reliability based on the assessment of CICS academic stakeholders and students.

1.6 Significance of the Study

This research is conducted to provide data-driven insights and essential information on the possible existence of curriculum gaps in the College of Information and Computer Studies at De La Salle University-Dasmariñas. The analysis of curriculum, course syllabi, and demanded skills from job postings in the technological field will determine the competencies that may require further strengthening, as well as skills and concepts that need to be introduced to meet the industry demands and expectations. The development of the system of this study is envisioned to have significance to the following:

CICS students. The study will benefit students at De La Salle University-Dasmariñas, specifically in the College of Information and Computer Studies, in terms of finding curriculum deficiencies with direct influence toward their preparedness for careers in the fast-changing technology sector. In addition, TechGap will also provide personalized skill roadmaps that guide them in addressing gaps not fully covered in their courses. Through recommended learning resources and career-aligned skill paths, the system allows the students to proactively enhance their competencies, enabling them to become more prepared, competitive, and adaptable to the rapidly evolving technology sector.

CICS faculty and academic administrators. The study will provide faculty members, as well as key academic decision-makers such as the Dean, Associate Dean,

and Department Chairs, with data-driven, prescriptive insights to support the evaluation and improvement of course offerings. By identifying specific skill gaps and recommending targeted curriculum revisions, the system assists administrators as well as the educators in integrating relevant technologies and concepts, refining teaching strategies, and strengthening the alignment between course content and industry expectations.

Industry partners and employers. The study provides value to industry stakeholders by promoting closer alignment between academic preparation and workforce requirements. By ensuring that graduates possess relevant and up-to-date skills, TechGap helps reduce skill mismatches, enabling the employers to have greater confidence in the competencies of graduates. This allows employers to recognize that graduates from CICS DLSU-D possess industry-relevant skills and are adequately prepared for professional roles, making them a reliable and job-ready talent pool that can be readily integrated into the workforce.

DLSU-D. The results of this study will serve as support for the university in making informed, strategic decisions regarding curriculum development and program enhancement. Through continuous assessment and simulation of proposed curriculum changes, the university can ensure that its academic programs remain relevant, up-to-date, and aligned with current industry standards. This ensures that DLSU-D continues to deliver high-quality education that effectively prepares students for

professional practice, while maintaining and strengthening its reputation as an institution that produces competent, industry-ready graduates.

Other researchers. The study benefits other researchers by introducing a data-driven approach to understanding curriculum–industry alignment. It offers insights into how skill requirements are analyzed against academic content, which may support the development of similar studies or the improvement of existing methodologies. As a result, the study contributes to further research on curriculum evaluation and alignment, as well as the exploration of data-driven approaches in academic program development.

1.7 Scope and Limitations

This study covers the development and evaluation of TechGap, a web-based curriculum gap analysis and decision-support system intended for the College of Information and Computer Studies (CICS) at De La Salle University-Dasmariñas. The primary objective is to analyze the alignment between the official CICS curriculum for the Bachelor of Science in Computer Science and Bachelor of Science in Information Technology programs and current technology industry skill requirements. To achieve this, the system applies a structured machine learning pipeline featuring natural language processing, K-Means clustering, and a fine-tuned Siamese Network to evaluate the semantic relationship between academic content and industry demands.

The academic reference consists of officially documented PDF syllabi of CICS specialization track subjects manually parsed into structured data files, with General Education courses strictly excluded from the skill matching analysis. Industry skill requirements are sourced directly from JobStreet, utilizing a dataset of approximately 5,000 job postings and competency descriptions relevant to computing roles. For academic stakeholders, TechGap equips curriculum planners with data-driven recommendations that strictly comply with CHED CMO No. 25, Series of 2015 guidelines by ensuring suggestions remain additive and protect mandated competencies. Additionally, a student-facing module allows students to visualize their personal alignment scores and receive targeted upskilling recommendations based solely on the official CICS courses they have completed.

Despite its specific capabilities, the study operates within several defined boundaries. The scope is strictly confined to the CICS department and does not extend to other colleges or institutions. To ensure high data fidelity and structural consistency for the semantic analysis, the system utilizes structured curriculum datasets parsed from official syllabus PDFs. This methodological approach prioritizes the accuracy of extracted course learning outcomes and technical modules by utilizing pre-validated data inputs to ensure the reliability of the semantic matching engine. Furthermore, all informal learning experiences such as internships, seminars, on-the-job training, external certifications, and extracurricular activities are entirely excluded, meaning a student's calculated alignment score relies exclusively on officially documented syllabi. The system also assumes that a student passing a course equates to full mastery of its respective skills without integrating technical assessments to verify actual knowledge retention.

While the system architecture is designed for dynamic data acquisition, the evaluation phase of this study utilizes a controlled, processed dataset to ensure a verifiable testing environment. Expanded data collection and additional continuous scraping are scheduled for the subsequent phase of the research project in Thesis 2. This limitation may affect the completeness of the identified skill demands. Ultimately, TechGap operates solely as an advisory tool that does not automatically modify institutional databases or enforce curriculum changes, leaving all final curriculum adjustments entirely to the discretion of academic administrators.